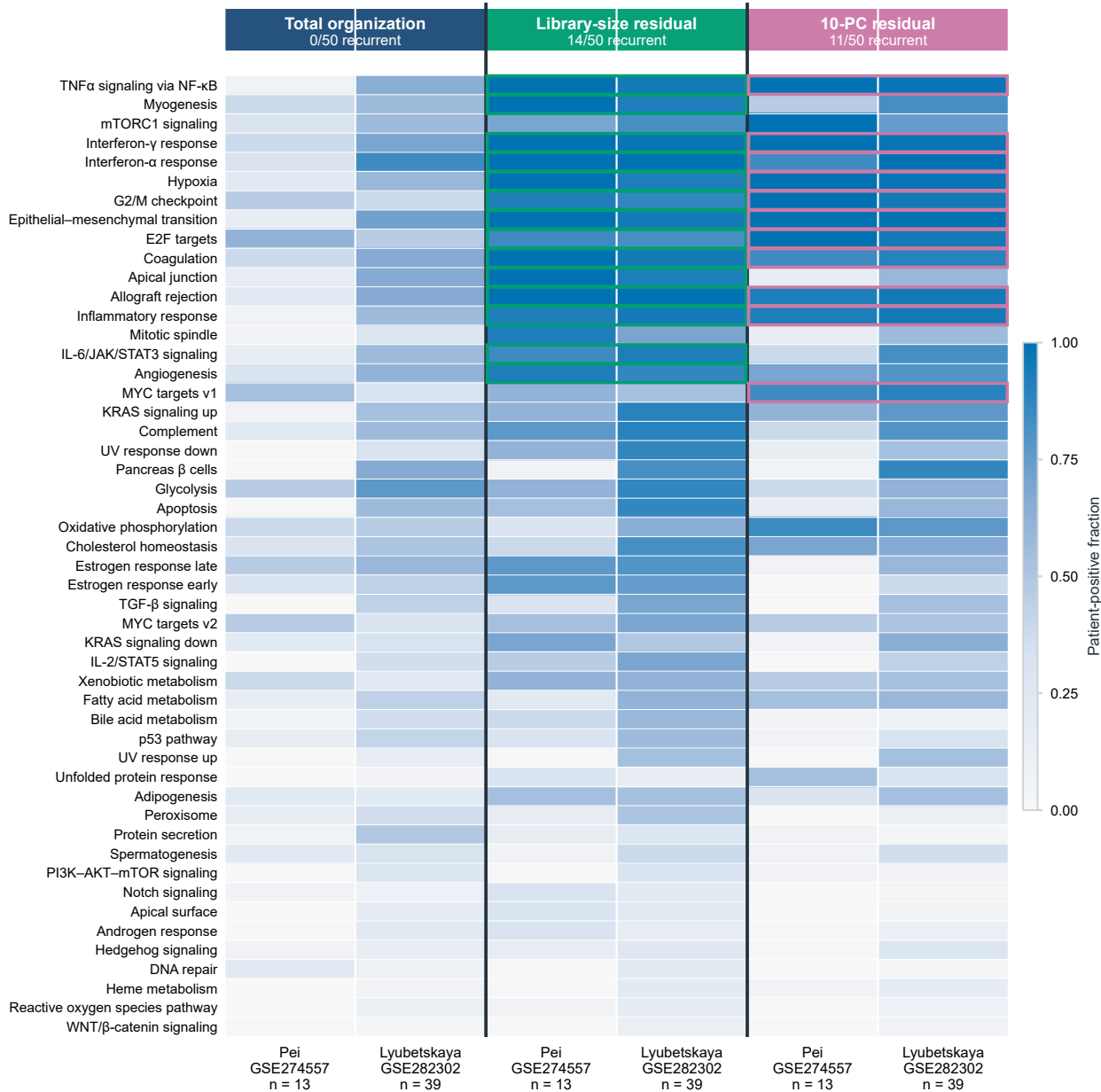


A Patient-positive fractions across 50 Hallmark pathways

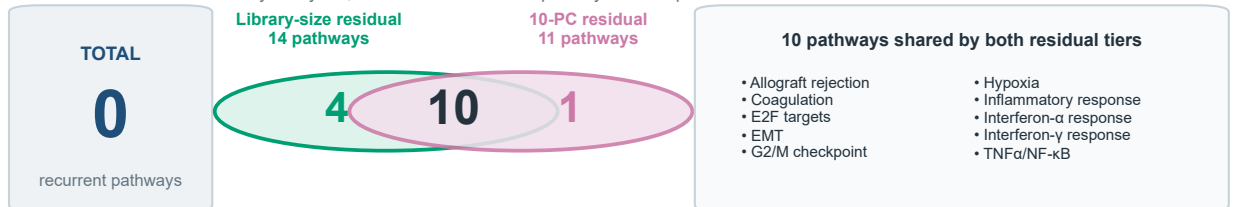
Outlined row-pairs meet the fixed $\geq 80\%$ recurrence rule in both cohorts



B

Recurrence summary and overlap

Residual tiers are sensitivity analyses, not substitutes for the primary total-map estimand



Key qualification: recurrence after residualization is recurrence of residual spatial organization, not of the unadjusted activity map.